

CSCI 4730/6730 Operating Systems

Programming Assignment #1 - Multi-process and IPC

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1. Multi-process and IPC Design

The parent opens the file and gets its size.

The file is divided into parts. Each part has an offset and a size.

The parent creates multiple child processes using `fork()`. Each child works on only its assigned part of the file.

Each child opens the file and calls `word_count()` with its assigned offset and size.

Each child has a separate ordinary pipe. The child writes its `count_t` result to the pipe. The parent reads the result from the pipe.

The parent adds each successful child result to the final total.

The parent creates all initial children before it starts waiting. This allows the children to run in parallel.

2. Crash Handling

The crash rate comes from the third command-line argument.

The `word_count()` function may call `abort()` to simulate a child crash.

The parent uses `waitpid()` to wait for each child. It checks whether the child exited successfully using `WIFEXITED()` and `WEXITSTATUS()`. Any non-successful child status is treated as a failed job.

If the child finishes successfully, the parent reads its result and adds it to the final total.

If the child fails, the parent does not add a result from that child.

The parent closes the old pipe, creates a new pipe, and creates a replacement child.

The replacement child repeats the same file range using the same offset and size.

This continues until the job succeeds or `MAX_FORK` is reached.